INCH-POUND

MIL-T-27/214A
20 April 1993
SUPERSEDING
MIL-T-27/214
6 March 1978

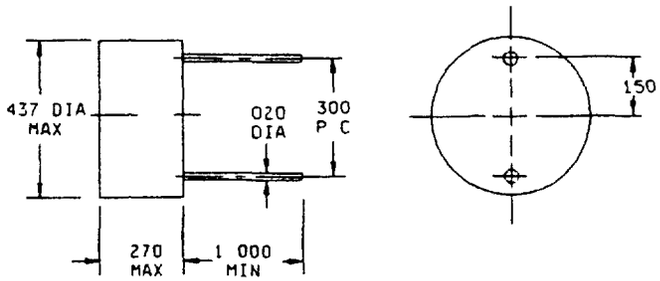
MILITARY SPECIFICATION SHEET

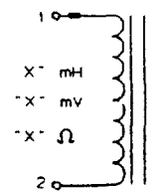
TRANSFORMERS AND INDUCTORS (AUDIO, POWER, AND HIGH-POWER PULSE) INDUCTORS, AUDIO FREQUENCY, TF5S20ZZ

(A) Inactive for new design after the date of this specification

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-T-27





WV 250 V
ALTITUDE 70 000 FT MAX
(SEE TABLE I FOR VALUE OF X)

CIRCUIT DIAGRAM AND MARKING

Inches	mm
.010	0.25
.020	0.51
150	3.81
.270	6.86
.300	7.62
437	11 10
1.000	25 40

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for information only
3. Tolerance on terminal position dimensions is ± 010
4. Marking shall be on the side of the case opposite terminals
5. Electrical values shall be marked as specified in table I, as applicable

FIGURE 1 Dimensions and configurations

(A) denotes changes

FSC 5950

DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

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REQUIREMENTS. (When numbers in parentheses, i e , (1-2) are used, they indicate the winding and the extreme terminals of the windings)

Electrical ratings See table I

Working voltage 250 volts

Design and construction

Dimensions and configuration See figure 1

Duty cycle Continuous

Case Encapsulated.

Material Epoxy

Terminals Tinned copper, 24 AWG

Terminal length 1 inch maximum

Weight 2 grams maximum

Operating temperature range -55°C to +130°C

Altitude 70,000 feet maximum

Terminal strength MIL-STD-202, method 211, test condition A, 2 pounds

Dielectric withstanding voltage

At sea level 1,000 volts rms
At reduced barometric pressure 300 volts rms

Electrical characteristics (see table I) Power level 0 5 watt maximum

Temperature rise 40°C at an ambient temperature of 90°C

Vibration (high frequency) MIL-STD-202, method 204, test condition D

(A) Quality assurance provisions

Qualification inspection Not applicable for this specification sheet

Quality conformance inspection Groups A and B tests of MIL-T-27 shall be applicable

Marking location See figure 1

Part or Identifying Number (PIN) M27/214-(dash number from table I)

Supersession data

MIL-T-27/214 supersedes MIL-T-94(USAF) which is inactive for new design

Superseding PIN'S

M27/214-08, M27/214-11 and M27/214-14 supersedes M27/94-01, M27/94-02 and M27/94-03, respectively

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TABLE I Electrical ratings

Dash no	Inductance (mH) (1-2)	Voltage (mV) (1-2)	Self-resonant frequency (MHz) (min)	DC resistance (ohms) (1-2) (max)	Temperature coefficient (-55°C to +130°C	Quality factor (min)	Distrib capacitance (pF) (max)
01	0 05 ±2%	11 3	9 000	1 89	±50 PPM	37 at 300 kHz	6
02	0 10 ±2%	16 0	6 000	2 64	±50 PPM	45 at 250 kHz	7
03	0 33 ±2%	29 2	2 800	7 20	±50 PPM	45 at 250 kHz	10
04	0 47 ±2%	34 9	2 300	10 00	±50 PPM	45 at 250 kHz	10
05	0 47 ±2%	13 8	2 200	2 30	±1%	20 at 20 kHz	11
06	1 00 ±2%	50 6	1 500	24 50	±50 PPM	23 at 400 kHz	15
07	1 00 ±2%	20 1	1 100	3 94	±1%	30 at 20 kHz	21
08	1 00 ±2%	13 7	1 800	2 30	±1%	40 at 100 kHz	6
09	3 30 ±1%	92 2	0 630	76 00	±50 PPM	23 at 100 kHz	19
10	3 30 ±1%	36 7	0 720	9 00	±1%	30 at 20 kHz	15
11	3 30 ±1%	25 0	0 860	5 50	±1%	40 at 100 kHz	9
12	10 00 ±1%	160 0	0 310	239 00	±50 PPM	23 at 100 kHz	26
13	10 00 ±1%	43 8	0 390	17 10	±1%	27 at 8 kHz	17
14	10 00 ±1%	65 0	0 475	26 00	±0 25%	60 at 100 kHz	15
15	33 00 ±1%	116 0	0 160	114 00	±1%	28 at 20 kHz	30
16	33 00 ±1%	74.9	0 190	50 20	±1%	14 at 5 kHz	21
17	100 00 ±1%	201 2	0 060	396 00	±1%	14 at 10 kHz	70
18	100 00 ±1%	138 6	0 090	207 00	±1%	14 at 5 kHz	31
19	250.00 ±1%	219 0	0 049	435 00	±1%	14 at 5 kHz	42

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CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - OS, SH
Air Force - 17, 99
DLA - ES

User activities
Army - AR, WC
Navy - AS, MC
Air Force - 19

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5950-0815-23)

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